Jasmin Bait

QUESTION 1: ESSAY

(a) According to most correlations, what are the two **variables** that can change external mass transfer coefficient?
(b) How can the mass transfer resistance be reduced?

Given Answer: a. concentration of the bulk and concentration of the catalyst surface
b. mass transfer resistance can be reduced by increasing temperature

Score: 15

QUESTION 2: ESSAY

1.

In a heterogeneous catalytic reaction, the measured order of reaction was different from the actual one. In fact, the $n_{actual} = 2.n_{measured} - 1$. Explain why the actual reaction-order was different.

Given Answer: [None Given]

Score : 0

QUESTION 3: ESSAY

1.

An irreversible, reaction-limited, dual-site reaction $A + B \rightarrow C + D$ is taking place on a catalyst surface. When the P_D is doubled, the rate of reaction stays the same, while when P_C was doubled, the rate reduced. If, A and B both adsorb on the catalyst surface, write the most likely reaction rate equation.

Given Answer: [None Given]

Score:0

QUESTION 4: ESSAY

1.

For a reaction A → B on a catalyst surface, the process is mass-transfer limited. The concentration in the bulk is C_{Ab} and on the surface is C_{AS}. The rate of surface reaction is given by $r_s = k_s C_{AS}^2/(1+K_A C_{AS})^2$. If the mass-transfer coefficient is kc, what is the overall rate of reaction?

Given Answer: $r = k_C*(C_{AB}-C_{AS})$

Score:20